

Project: Mathematics IV

Early-Stage Epidemic Control: Linear Systems and Laplace Transforms

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Abstract

This project investigates the dynamics of an infectious disease outbreak and the impact of quarantine interventions. To apply analytical methods, we restrict the classical SIR model to its early-stage linear approximation. We evaluate the system using fundamental matrices and eigenvalues to determine outbreak thresholds linked to the basic reproduction number (R_0). Furthermore, we apply the Laplace Transform method to mathematically model a delayed quarantine rollout using a Heaviside step function.

Introduction

While standard epidemiological models (like the SIR model) rely on highly non-linear differential equations, the initial phase of an outbreak can be closely approximated using linear systems. During this early stage, the susceptible population $S(t)$ is approximately equal to the total population N . This project utilizes matrix methods and Laplace transforms to model how varying public health strategies control the spread of infection.

Linearized System with Quarantine Control

Standard epidemiological models rely on non-linear dynamics, which naturally bound infection growth as the susceptible population depletes. In a standard model, the rate of new infections relies on the product of susceptible and infected populations, $\beta S(t)I(t)$ (where β is the disease transmission rate). The rate of natural recovery is $\gamma I(t)$ (where γ is the recovery rate). This gives the non-linear ordinary differential equation:

$$I'(t) = \beta S(t)I(t) - \gamma I(t) \tag{1}$$

To rigorously apply analytical methods, we restrict our model to the early-stage approximation where the susceptible population is nearly the entire population ($S(t) \approx N$). Substituting N into the equation and factoring yields a linear relationship:

$$I'(t) = (\beta N - \gamma)I(t) \quad (2)$$

We define the natural growth rate of the infection as $\alpha = \beta N - \gamma$.

To prevent physically impossible unbounded growth in this linear regime, our control variable must act directly on the infected pool. We define $I(t)$ as the active infected population and $Q(t)$ as the quarantined population. We model a delayed public health response that removes a constant capacity of Q_0 individuals per day into quarantine, starting at day T . This constant rate reflects real-world logistical constraints, such as a fixed maximum number of available hospital beds or daily testing limits. Quarantined individuals recover at a rate δ .

This forms a non-homogeneous system of linear ODEs:

$$I'(t) = \alpha I(t) - Q_0 H(t - T) \quad (3)$$

$$Q'(t) = Q_0 H(t - T) - \delta Q(t) \quad (4)$$

Because the system is linear and the forcing function $H(t - T)$ is piecewise continuous, Picard's Existence and Uniqueness Theorem guarantees a unique solution for any given initial condition, which we define as $I(0) = I_0$ for the infected population.

Matrix Methods and Homogeneous Stability

To understand the system's natural behavior before any intervention ($t < T$), we evaluate the homogeneous portion of the system. During this period, the quarantine is inactive, meaning the Heaviside step function evaluates to zero ($H(t - T) = 0$). Substituting this into our non-homogeneous equations yields a decoupled system of linear ODEs:

$$I'(t) = \alpha I(t) \quad (5)$$

$$Q'(t) = -\delta Q(t) \quad (6)$$

We can express this homogeneous system in matrix form $\mathbf{X}' = \mathbf{A}\mathbf{X}$:

$$\begin{pmatrix} I' \\ Q' \end{pmatrix} = \begin{pmatrix} \alpha & 0 \\ 0 & -\delta \end{pmatrix} \begin{pmatrix} I \\ Q \end{pmatrix} \quad (7)$$

Because A is a diagonal matrix, its eigenvalues are explicitly $\lambda_1 = \alpha$ and $\lambda_2 = -\delta$. The fundamental matrix is explicitly given by:

$$\Phi(t) = \begin{pmatrix} e^{\alpha t} & 0 \\ 0 & e^{-\delta t} \end{pmatrix} \quad (8)$$

This dictates that if $\alpha > 0$, the $e^{\alpha t}$ term forces the infection to exhibit unbounded exponential growth. We can rewrite the condition $\alpha > 0$ as $\beta N - \gamma > 0$, or $\frac{\beta N}{\gamma} > 1$. This ratio is exactly the basic reproduction number, R_0 . If $R_0 > 1$, mathematical intervention is strictly necessary.

Modeling Delayed Quarantine via Laplace Transforms

To find the exact analytical solution for the discontinuous forcing function, we apply the Laplace Transform ($\mathcal{L}$), where s is the complex frequency variable, to the infected population equation:

$$\mathcal{L}\{I'(t)\} = \alpha\mathcal{L}\{I(t)\} - Q_0\mathcal{L}\{H(t-T)\} \quad (9)$$

Applying standard transform tables (specifically the derivative rule and the shifting theorem for the Heaviside step function) and the initial condition $I(0) = I_0$:

$$sI(s) - I_0 = \alpha I(s) - Q_0 \frac{e^{-sT}}{s} \quad (10)$$

We isolate $I(s)$ algebraically to find the exact solution in the frequency domain:

$$I(s)(s - \alpha) = I_0 - Q_0 \frac{e^{-sT}}{s} \quad (11)$$

$$I(s) = \frac{I_0}{s - \alpha} - \frac{Q_0 e^{-sT}}{s(s - \alpha)} \quad (12)$$

To perform the inverse Laplace transform ($\mathcal{L}^{-1}$), we apply partial fraction decomposition to the second term: $\frac{1}{s(s-\alpha)} = \frac{1/\alpha}{s-\alpha} - \frac{1/\alpha}{s}$. Substituting this back into the equation gives:

$$I(s) = \frac{I_0}{s - \alpha} - \frac{Q_0}{\alpha} \left(\frac{e^{-sT}}{s - \alpha} - \frac{e^{-sT}}{s} \right) \quad (13)$$

Applying the Second Shifting Theorem yields the exact closed-form solution in the time domain:

$$I(t) = I_0 e^{\alpha t} - \frac{Q_0}{\alpha} (e^{\alpha(t-T)} - 1) H(t-T) \quad (14)$$

Analytical Threshold for Epidemic Collapse

To determine the necessary quarantine capacity, we analyze the system's behavior after the intervention has begun (for $t \geq T$). Under this condition, the Heaviside step function activates, meaning $H(t-T) = 1$. We substitute this into our time-domain solution:

$$I(t) = I_0 e^{\alpha t} - \frac{Q_0}{\alpha} (e^{\alpha(t-T)} - 1) \quad (15)$$

Next, we expand the exponent $e^{\alpha(t-T)}$ as $e^{\alpha t} e^{-\alpha T}$:

$$I(t) = I_0 e^{\alpha t} - \frac{Q_0}{\alpha} (e^{\alpha t} e^{-\alpha T} - 1) \quad (16)$$

Distributing the fraction $-\frac{Q_0}{\alpha}$ yields:

$$I(t) = I_0 e^{\alpha t} - \frac{Q_0}{\alpha} e^{\alpha t} e^{-\alpha T} + \frac{Q_0}{\alpha} \quad (17)$$

Finally, we factor out the common $e^{\alpha t}$ term to isolate the coefficient driving the exponential growth:

$$I(t) = e^{\alpha t} \left(I_0 - \frac{Q_0}{\alpha} e^{-\alpha T} \right) + \frac{Q_0}{\alpha} \quad (18)$$

To prevent exponential explosion, the coefficient of $e^{\alpha t}$ must be forced negative. Therefore, we analytically derive the minimum required quarantine capacity to collapse the epidemic:

$$Q_0 > I_0 \alpha e^{\alpha T} \quad (19)$$

If this threshold is met, the infection curve peaks strictly at $t = T$ and rapidly decays to zero.

Sensitivity Analysis

Using the derived threshold formula, we can analyze the impact of logistical constraints:

- **Delayed Response ($T \uparrow$):** Since the delay T acts in the exponent, any increase in intervention time requires an exponentially larger Q_0 to arrest the outbreak. For a fixed capacity, increasing T eventually violates the threshold.
- **Lower Capacity ($Q_0 \downarrow$):** If Q_0 drops below the threshold, the coefficient of $e^{\alpha t}$ remains positive. The growth rate slows but remains exponential, failing to collapse the active infection curve.

Numerical Simulation

We simulate the non-homogeneous system computationally with initial infected $I_0 = 0.01$ and growth rate $\alpha = 0.15$. The outbreak grows unchecked until $T = 15$ days, reaching $I(15) \approx 0.095$. According to our threshold, the quarantine capacity must exceed $Q_0 > 0.014$. We implement a successful intervention ($Q_0 = 0.03$) and a weak intervention ($Q_0 = 0.012$, below threshold) to demonstrate the boundary condition.

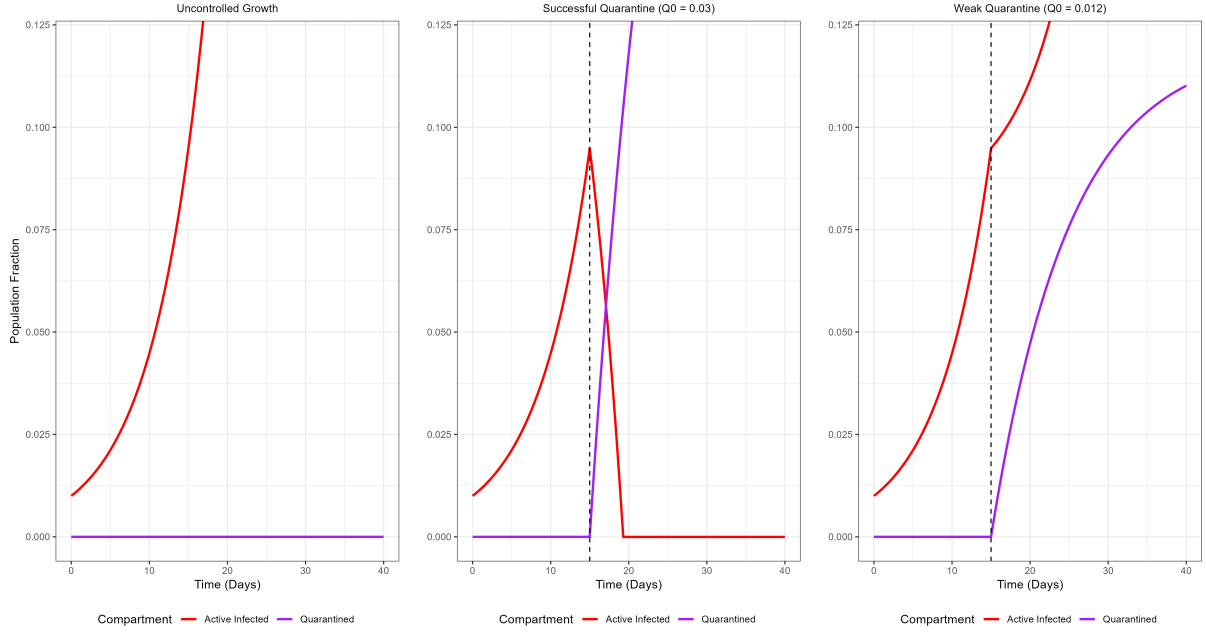


Figure 1: Simulation of the linear non-homogeneous system. The successful strategy ($Q_0 = 0.03$) forces the exponential coefficient negative. The weak strategy ($Q_0 = 0.012$) slows the spread but fails to invert the exponential growth.

Limitations of the Linear Approximation: The Saturation Effect

A fundamental limitation of this linearized model is the assumption that the susceptible population remains constant ($S \approx N$). In a full non-linear SIR model, the infection rate relies on the product $S(t)I(t)$. As infections rise, $S(t)$ depletes, introducing a natural saturation effect that slows transmission. By treating S as constant, our linear system inherently overestimates growth and will predict boundless expansion towards infinity unless manually controlled via Q_0 . In reality, the epidemic would eventually peak autonomously.

Conclusion

By utilizing a non-homogeneous linear system, we modeled a realistic epidemic response. Matrix eigenvalue analysis confirmed initial instability linked to R_0 , while Laplace transforms provided an exact algebraic method to calculate the required intervention required to invert exponential growth. Though limited by the absence of non-linear saturation effects, this approximation yields rigorous, closed-form thresholds vital for early-stage public health planning.

Appendix: R Code for Numerical Simulation

The following R script explicitly utilizes a step-function ('ifelse') to simulate the delayed control strategies, comparing outcomes above and below the derived capacity threshold.

```
library(deSolve)
library(ggplot2)
library(tidyr)
library(gridExtra)

# 1. Define the Non-Homogeneous Linear System
linear_quarantine <- function(time, state, parameters) {
  with(as.list(c(state, parameters)), {

    # Heaviside step function: Quarantine capacity Q0 activates at day
    # T
    current_Q <- ifelse(time >= delay_T, Q0, 0)

    # Linear ODEs from the analytical derivation
    dI <- alpha * I - current_Q
    dQ <- current_Q - delta * Q

    # Prevent populations from mathematically dropping below zero
    dI <- ifelse(I <= 0 & dI < 0, 0, dI)
    dQ <- ifelse(Q <= 0 & dQ < 0, 0, dQ)

    return(list(c(dI, dQ)))
  })
}

# 2. Initial conditions and restricted time grid
init_state <- c(I = 0.01, Q = 0.0)
times <- seq(0, 40, by = 0.1)

# 3. Parameters
params_no_control <- c(alpha = 0.15, delta = 0.1, Q0 = 0.0, delay_T =
  0)
params_control <- c(alpha = 0.15, delta = 0.1, Q0 = 0.03, delay_T =
  15)
params_weak <- c(alpha = 0.15, delta = 0.1, Q0 = 0.012, delay_T =
  15) # Below threshold

# 4. Solve the ODEs
out_no <- as.data.frame(ode(y = init_state, times = times, func =
  linear_quarantine, parms = params_no_control))
out_yes <- as.data.frame(ode(y = init_state, times = times, func =
  linear_quarantine, parms = params_control))
out_weak <- as.data.frame(ode(y = init_state, times = times, func =
  linear_quarantine, parms = params_weak))

out_no_long <- pivot_longer(out_no, cols = c("I", "Q"), names_to = "
  Compartment", values_to = "Fraction")
out_yes_long <- pivot_longer(out_yes, cols = c("I", "Q"), names_to = "
  Compartment", values_to = "Fraction")
out_weak_long <- pivot_longer(out_weak, cols = c("I", "Q"), names_to =
  "Compartment", values_to = "Fraction")

# 5. Plotting
```

```

line_colors <- c("I" = "red", "Q" = "purple")
shared_theme <- theme_bw() + theme(legend.position = "bottom", plot.
  title = element_text(hjust = 0.5, size=10))

plot1 <- ggplot(out_no_long, aes(x = time, y = Fraction, color =
  Compartment)) +
  geom_line(linewidth = 1) +
  scale_color_manual(values = line_colors, labels = c("I" = "Active
    Infected", "Q" = "Quarantined")) +
  labs(title = "Uncontrolled Growth", x = "Time (Days)", y = "
    Population Fraction") +
  coord_cartesian(ylim = c(0, 0.12)) + shared_theme

plot2 <- ggplot(out_yes_long, aes(x = time, y = Fraction, color =
  Compartment)) +
  geom_line(linewidth = 1) +
  geom_vline(xintercept = 15, linetype="dashed", color = "black") +
  scale_color_manual(values = line_colors, labels = c("I" = "Active
    Infected", "Q" = "Quarantined")) +
  labs(title = "Successful Quarantine (Q0 = 0.03)", x = "Time (Days)",
    y = "") +
  coord_cartesian(ylim = c(0, 0.12)) + shared_theme

plot3 <- ggplot(out_weak_long, aes(x = time, y = Fraction, color =
  Compartment)) +
  geom_line(linewidth = 1) +
  geom_vline(xintercept = 15, linetype="dashed", color = "black") +
  scale_color_manual(values = line_colors, labels = c("I" = "Active
    Infected", "Q" = "Quarantined")) +
  labs(title = "Weak Quarantine (Q0 = 0.012)", x = "Time (Days)", y = "
    ") +
  coord_cartesian(ylim = c(0, 0.12)) + shared_theme

combined_plot <- grid.arrange(plot1, plot2, plot3, ncol = 3)
ggsave("graph_updated.png", plot = combined_plot, width = 15, height =
  5, dpi = 300, bg = "white")

```